

OUTLIER ANALYSIS

DV2627 – Advanced Machine Learning



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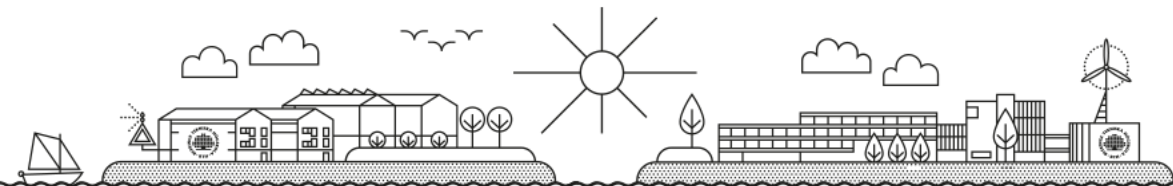
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OUTLINE

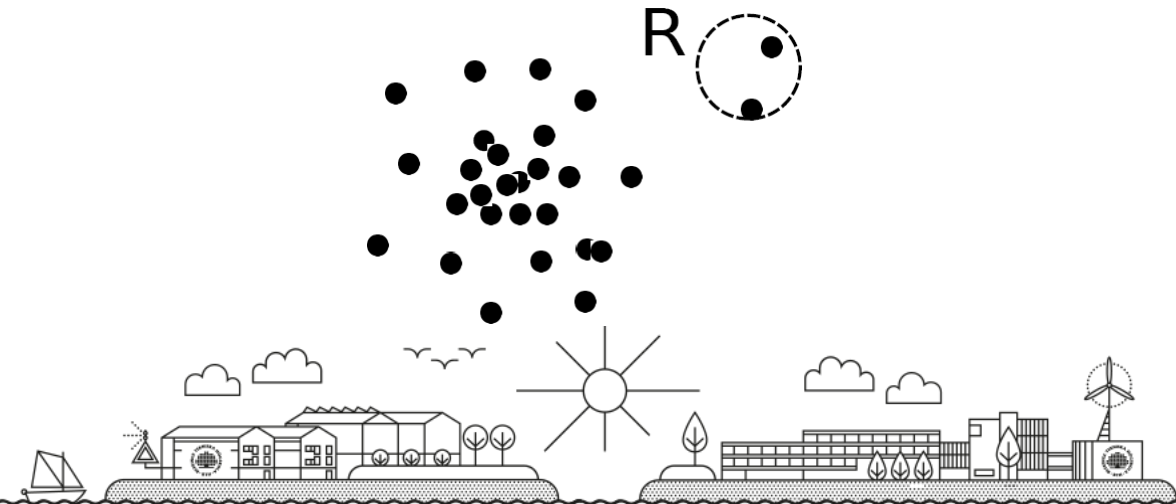


- **Outlier and Outlier Analysis**
- Outlier Detection Methods
- Statistical Approaches
- Proximity-Base Approaches
- Clustering-Base Approaches
- Classification Approaches
- Summary



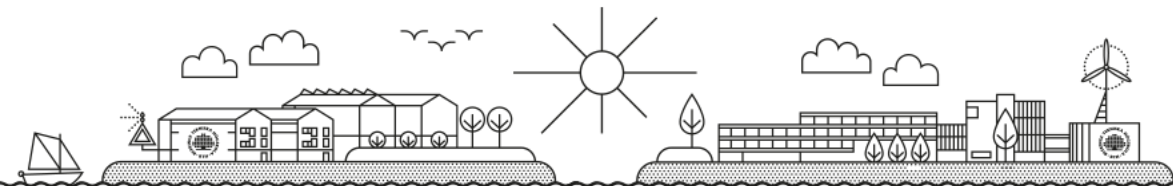
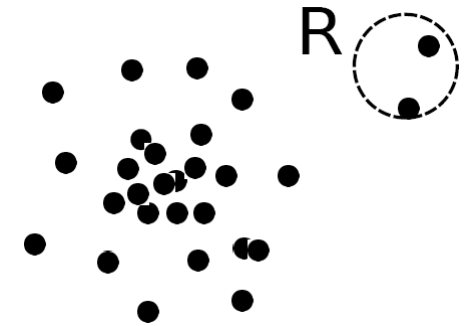
WHAT IS AN OUTLIER?

- **Outlier:** A data object that **deviates significantly** from most objects as if a different mechanism generated it
 - E.g., Unusual credit card purchases
- Outliers are different from the noise
 - Noise is random error or variance in a measured variable
 - Noise should be removed before outlier detection
- Outliers are interesting: they violate the mechanism that generates the normal data
- Outlier detection vs. *novelty detection*: early stage, outlier, but later merged into the model
- Applications:
 - Credit card fraud detection
 - Telecom fraud detection
 - Customer segmentation
 - Medical analysis



TYPES OF OUTLIERS (I)

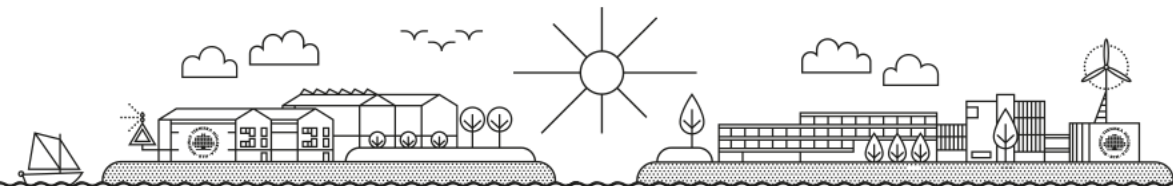
- Three kinds: *global*, *contextual*, and *collective* outliers
- **Global outlier** (or point anomaly)
 - An object is O_g if it significantly deviates from the rest of the data set
 - E.g., Intrusion detection in computer networks
 - **Issue:** Find an appropriate measurement of deviation



TYPES OF OUTLIERS (2)

- **Contextual outlier** (or *conditional outlier*)

- An object is O_c if it deviates significantly based on a selected context.
- E.g., -8°C in Karlskrona: outlier? (depending on summer or winter?)
- Attributes of data objects should be divided into two groups
 - **Contextual attributes:** defines the context, e.g., time and/or location
 - **Behavioral attributes:** characteristics of the object used in outlier evaluation, e.g., temperature
- It can be viewed as a generalization of *local outliers*—whose density significantly deviates from their local area
- **Issue:** How to define or formulate meaningful context? **Group-by**



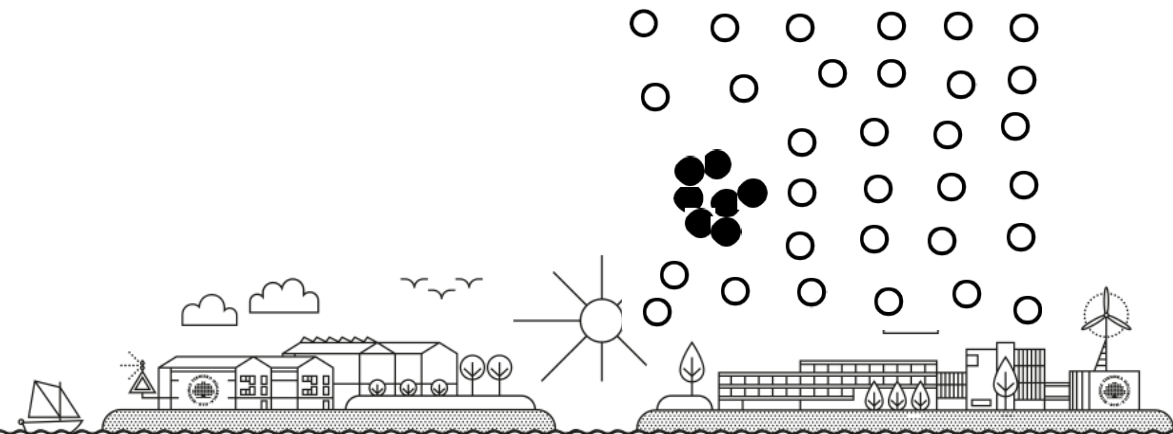
TYPES OF OUTLIERS (3)

○ Collective Outliers

- A subset of data objects *collectively* deviates significantly from the whole data set, even if the individual data objects may not be outliers
- E.g., intrusion detection:
 - When several computers keep sending denial-of-service packages to each other
- Detection of collective outliers
 - Consider not only the **behavior** of individual objects but also that of groups of objects
 - Need to have **background knowledge** on the relationship among data objects, such as a **distance or similarity measure** on objects.

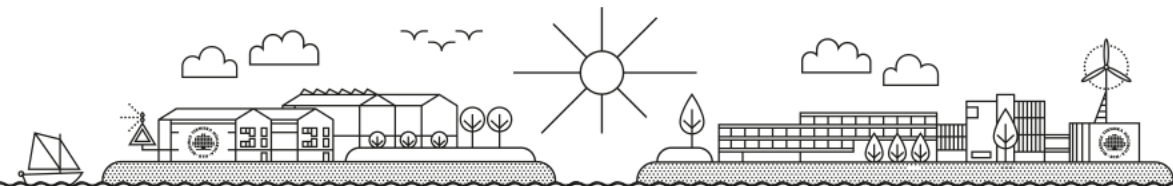
○ A data set may have multiple types of outlier

○ One object may belong to more than one type of outlier



CHALLENGES OF OUTLIER DETECTION

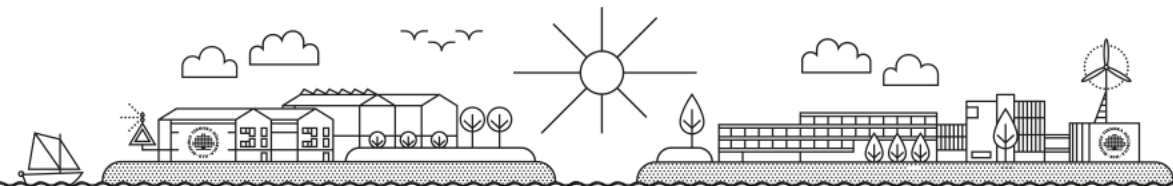
- Modeling normal objects and outliers properly
 - Hard to enumerate all possible normal behaviors in an application
 - The border between normal and outlier objects is often a gray area
- Application-specific outlier detection
 - The choice of **distance measure** among objects and **the model of the relationship** among objects are often application-dependent
 - E.g., in health applications, a small deviation could be an outlier, while in marketing analysis, larger fluctuations
- Handling noise in outlier detection
 - Noise may distort the normal objects and blur the distinction between normal objects and outliers.
 - It may help hide outliers and reduce the effectiveness of outlier detection
- Understandability
 - Understand why these are outliers
 - Specify the degree of an outlier



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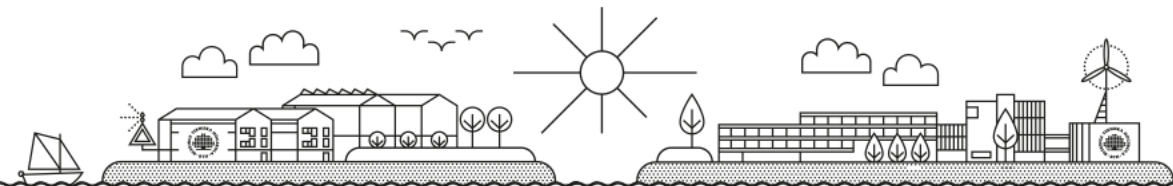


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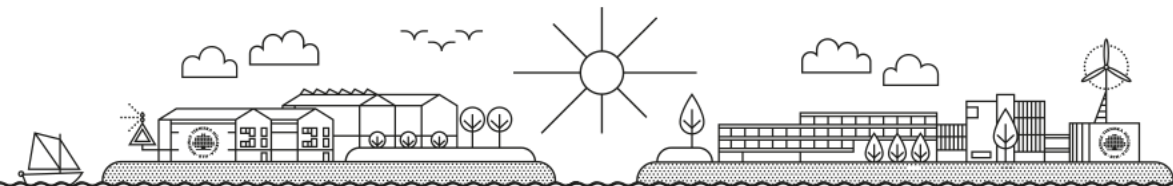
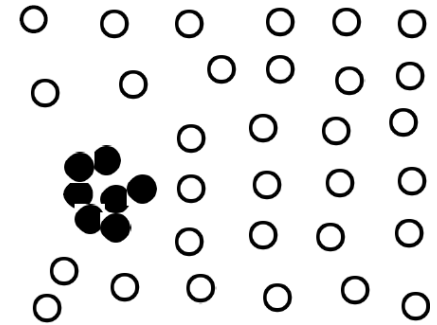
OUTLIER DETECTION (1): SUPERVISED METHODS

- Two ways to categorize outlier detection methods:
 - Based on whether user-labeled examples of outliers can be obtained:
 - Supervised, semi-supervised, and unsupervised methods
 - Based on assumptions about normal data and outliers:
 - Statistical, proximity-based, and clustering-based methods
- **Supervised Methods**
 - Modeling outlier detection as a classification problem
 - Samples examined by domain experts used for training and testing
 - Methods for learning a classifier for outlier detection effectively:
 - Model normal and outlying behavior
 - Challenges
 - Imbalanced classes
 - ✓ Boost the outlier class and make up some artificial outliers
 - Catch as many outliers as possible, i.e., **recall** is more important than **accuracy** (i.e., not mislabeling normal objects as outliers)



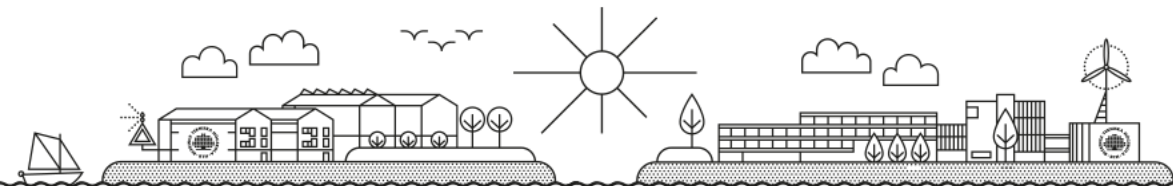
OUTLIER DETECTION (2): UNSUPERVISED METHODS

- Assume the normal objects are somewhat **clustered** into multiple groups, each having some distinct features
- An outlier is expected to be far away from any groups of normal objects
- **Weakness:** Cannot detect collective outliers effectively
 - Normal objects may not share any strong patterns, but the collective outliers may share high similarities in a small area
 - E.g., In some intrusion or virus detection, normal activities are diverse
 - Unsupervised methods may have a high false positive rate but still, miss many real outliers.
- Many clustering methods can be adapted for unsupervised methods
 - Find clusters, then outliers: not belonging to any cluster
 - **Problem 1:** Hard to distinguish noise from outliers
 - **Problem 2:** Costly since first clustering: but far fewer outliers than normal objects
 - Newer methods: tackle outliers directly, proximity-based



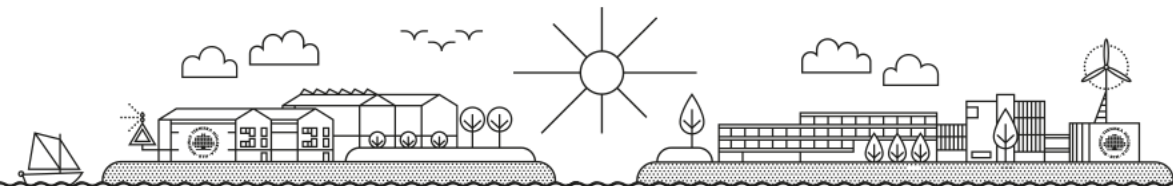
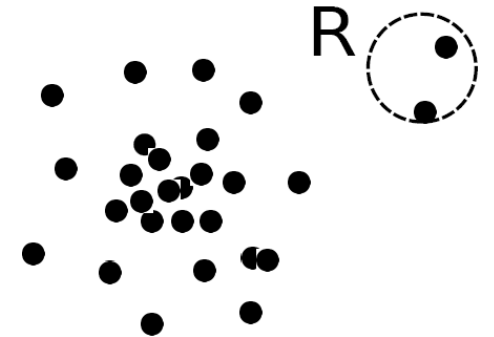
OUTLIER DETECTION (3): SEMI-SUPERVISED METHODS

- **Situation:** In many applications, the number of labeled data is often small: Labels could be on outliers only, normal objects only, or both
- If some labeled normal objects are available
 - Use the labeled examples and the proximate unlabeled objects to train a model for normal objects
 - Those not fitting the model of normal objects are detected as outliers
- If only some labeled outliers are available, a small number of labeled outliers many not cover the possible outliers well
 - To improve the quality of outlier detection, one can get help from models for normal objects learned from unsupervised methods



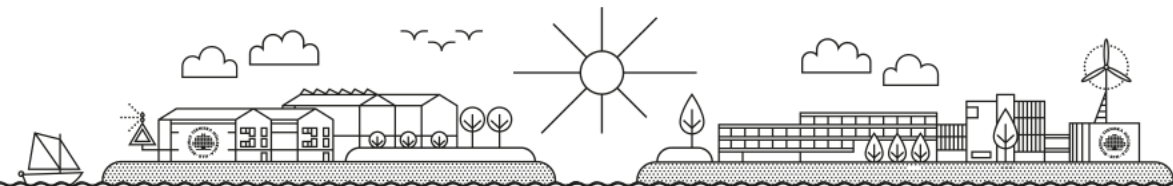
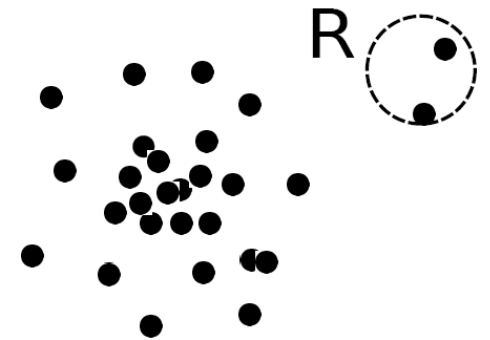
OUTLIER DETECTION (1): STATISTICAL METHODS

- Statistical methods (also known as model-based methods) assume that the normal data follow some statistical model (a stochastic model)
 - The data not following the model are outliers.
- E.g., First, use Gaussian distribution to model the normal data
 - For each object, y , in region R , estimate $g_D(y)$; the probability of y fits the Gaussian distribution
 - If $g_D(y)$ is very low, y is unlikely generated by the Gaussian model, thus an outlier
- Effectiveness of statistical methods
 - highly depends on whether the assumption of the statistical model holds in the real data
- Various statistical models
 - parametric vs. non-parametric



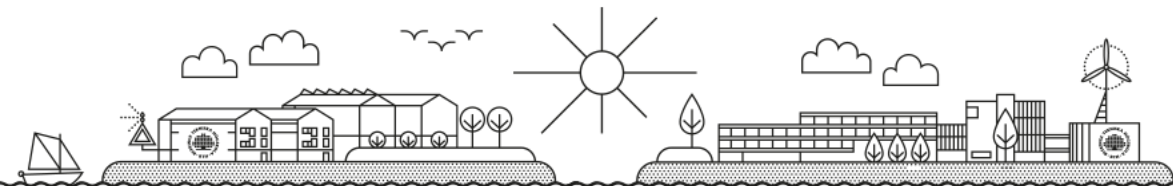
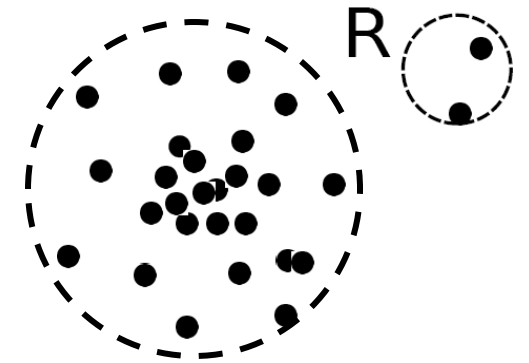
OUTLIER DETECTION (2): PROXIMITY-BASED METHODS

- An object is an outlier if the nearest neighbors of the object are far away, i.e., the **proximity** of the object **significantly deviates** from the proximity of most of the other objects in the same data set
- E.g., Model the proximity of an object using its 3 nearest neighbors
 - Objects in region R differ substantially from other objects in the data set.
 - Thus the objects in R are outliers
- The effectiveness of proximity-based methods highly relies on the proximity measure.
- In some applications, proximity or distance measures cannot be obtained easily.
- Often have difficulty finding a group of outliers that stay close to each other
- Two major types of proximity-based outlier detection
 - Distance-based vs. density-based



OUTLIER DETECTION (3): CLUSTERING-BASED METHODS

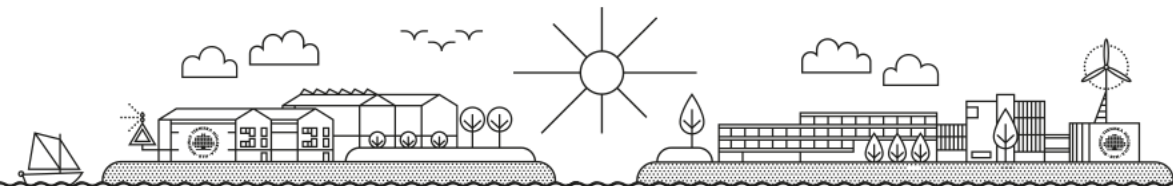
- Normal data belong to **large** and **dense** clusters, whereas outliers belong to **small** or sparse clusters, or do not belong to any clusters
- E.g., two clusters
 - All points not in R form a large cluster
 - The two points in R form a tiny cluster, thus are outliers
- Clustering is **expensive**: straightforward adaption of a clustering method for outlier detection can be costly and does **not scale up** well for large data sets



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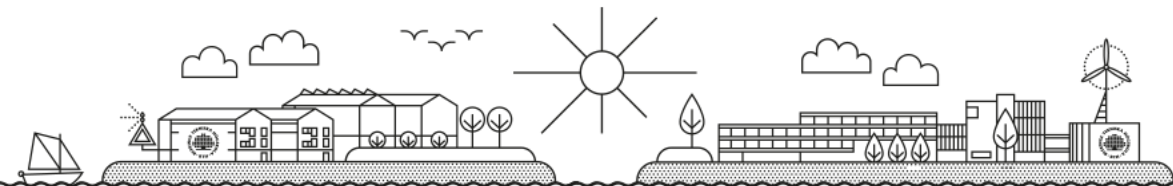


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STATISTICAL APPROACHES

- Statistical approaches assume that the objects in a data set are generated by a **stochastic** process (a generative model)
- **Idea:** learn a generative model fitting the given data set, and then identify the objects in **low probability** regions of the model as outliers
- **Methods are divided into two categories:** *parametric* vs. *non-parametric*
- Parametric method
 - Assumes that the normal data is generated by a parametric distribution with parameter θ
 - The probability density function of the parametric distribution $f(x, \theta)$ gives the probability that object x is generated by the distribution
 - The smaller this value, the more likely x is an outlier
- Non-parametric method
 - Not assume an apriori statistical model and determine the model from the input data
 - Not completely parameter-free but consider the number and nature of the parameters are flexible and not fixed in advance
 - Examples: histogram and kernel density estimation

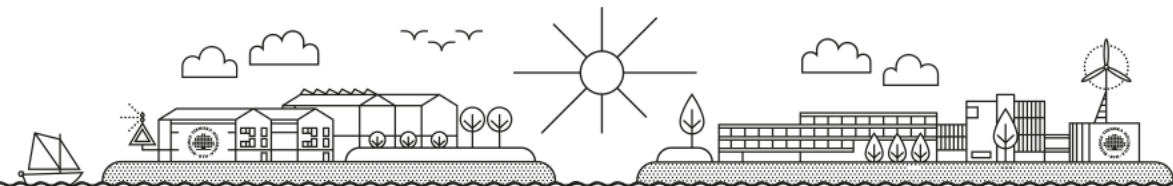


PARAMETRIC METHODS I: DETECTION UNIVARIATE OUTLIERS BASED ON NORMAL DISTRIBUTION

- Univariate data: A data set involving only one attribute or variable
- Often assume that data are generated from a normal distribution, learn the parameters from the input data, and identify the points with low probability as outliers
- E.g., Avg. temp.: {24.0, 28.9, 28.9, 29.0, 29.1, 29.1, 29.2, 29.2, 29.3, 29.4}

$$\hat{\mu} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

- For the above data with $n = 10$, we have $\hat{\mu} = 28.61$ and $\hat{\sigma} = \sqrt{2.29} = 1.51$
- Then $(28.61 - 24) / 1.51 = 3.04 > 3$, measurement with value 24 is an outlier since $\mu \pm 3\sigma$ region contains 99.7% data



PARAMETRIC METHODS I: THE GRUBB'S TEST

- Univariate outlier detection: The Grubb's test (maximum normed residual test) – another statistical method under the normal distribution

- For each object Y in a data set, compute its **absolute** Grubbs' test statistic considering mean, $\bar{Y}$, and standard deviation, s :

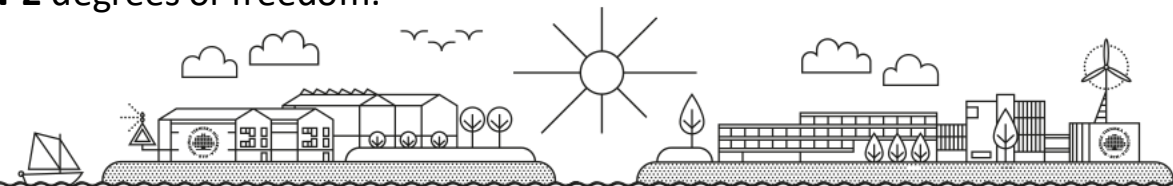
$$G = \frac{\max |Y_i - \bar{Y}|}{s}$$

- Y is an outlier if

$$G > \frac{(N-1)}{\sqrt{N}} \sqrt{\frac{(t_{\alpha/(2N), N-2})^2}{N-2 + (t_{\alpha/(2N), N-2})^2}}$$

Manually or using a lookup table for G

- Where $t_{\alpha/(2N), N-2}$ is the upper critical value of a **t-distribution** with **N-2** degrees of freedom.
- For a one-tailed test, replace $\alpha/(2N)$ with α/N .

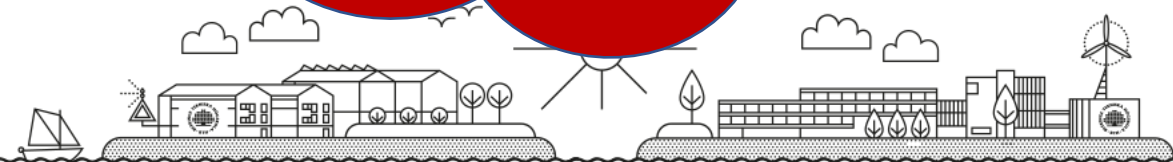


THE GRUBB'S TEST

- E.g., Sample = 199.31, 199.53, 200.19, 200.82, 201.92, 201.95, 202.18, **245.57**
- H_0 : there are no outliers in the data
- H_a : the maximum value is an outlier
- Test statistic:
 - **Mean** = 206.434, **Standard deviation**= 15.853
 - **Grabb's test**(245.57) = 2.469
 - Significance level: $\alpha = 0.05$
 - Critical value for an upper one-tailed test: 2.032
 - Critical region: Reject H_0 if **G** > 2.032

The Grubb's Test is for
identifying a single outlier.

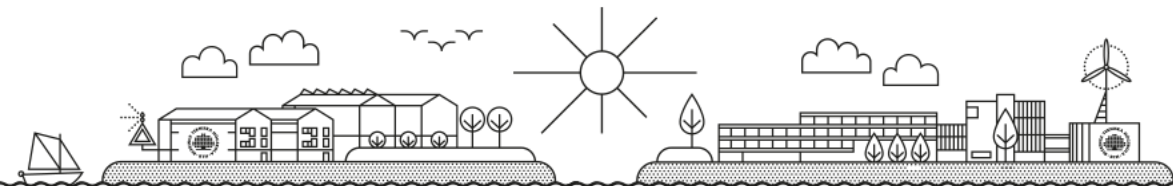
For more than one outlier use
Tietjen-Moore or Generalized
ESD Tests



PARAMETRIC METHODS II: DETECTION OF MULTIVARIATE OUTLIERS

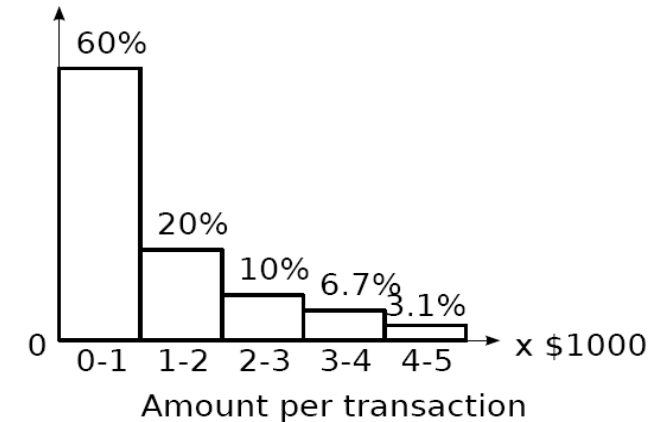
- Multivariate data: A data set involving two or more attributes or variables
- Transform the multivariate outlier detection task into a univariate outlier detection problem
- **Method 1.** Compute **Mahalanobis** distance, which is a measure of the distance between a sample point and a distribution.
 - Let $\bar{o}$ be the mean vector for a multivariate data set. Mahalanobis distance for an object o to $\bar{o}$ is $\text{MDist}(o, \bar{o}) = (o - \bar{o})^T S^{-1}(o - \bar{o})$ where S is the covariance matrix
 - Detect outliers on the transformed data
- **Method 2.** Use χ^2 -statistic (Chi-Square Statistic):
 - where E_i is the mean of the i th-dimension among all objects, and n is the dimensionality
 - If χ^2 -statistic is large with respect to a critical value from the Chi-Square table, then object o is an outlier

$$\chi^2 = \sum_{i=1}^n \frac{(o_i - E_i)^2}{E_i},$$



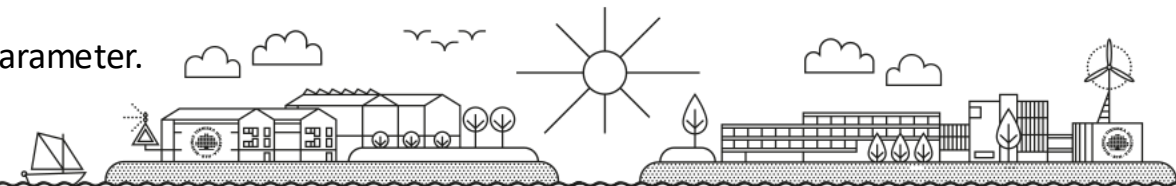
NON-PARAMETRIC METHODS: DETECTION OF OUTLIER USING HISTOGRAM

- The model of normal data is learned from the input data without any *a priori* structure.
- Often makes fewer assumptions about the data, and thus can be applied in more scenarios
- Outlier detection using histogram:
 - The histogram of purchase amounts in transactions
 - A transaction in the amount of \$7,500 is an outlier since only $100\% - (60\% + 20\% + 10\% + 6.7\% + 3.1\%) = 0.2\%$ transactions have an amount higher than \$5,000
- Problem: Hard to choose an **appropriate bin size** for the histogram
 - Too small bin size → normal objects in empty/rare bins, false positive
 - Too big bin size → outliers in some frequent bins, false negative
- **Solution:** Adopt **kernel density** estimation to estimate the probability density distribution of the data. If the estimated density function is high, the object is likely normal. Otherwise, it is likely an outlier.



$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right),$$

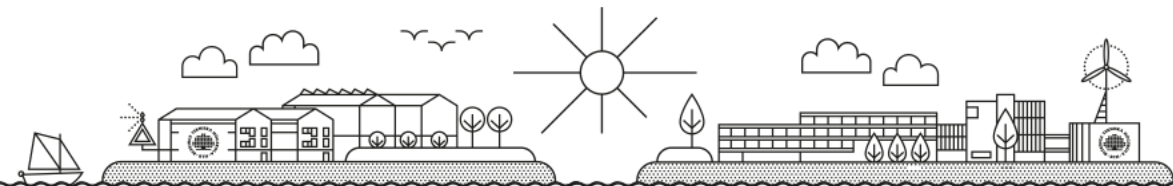
where $K()$ is a kernel and h is the bandwidth serving as a smoothing parameter. The density estimates at a point x within a group of points x_i ($i=1, \dots, n$)



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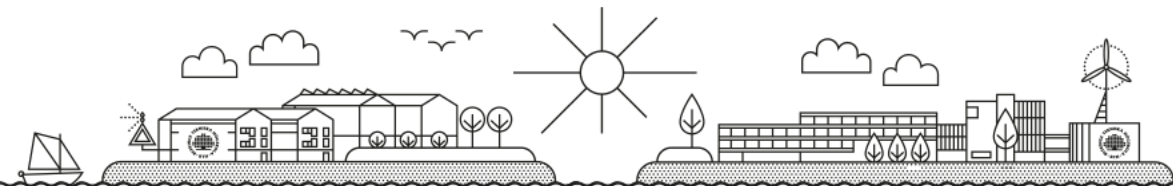


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PROXIMITY-BASED APPROACHES: DISTANCE-BASED

- **Intuition:** Objects that are far away from the others are outliers
- Assumption of proximity-based approach:
 - The proximity of an outlier **deviates significantly** from that of most of the others in the data set
- Two types of proximity-based outlier detection methods
 1. Distance-based outlier detection: An object o is an outlier if its neighborhood does not have **enough other points**
 2. Density-based outlier detection: An object o is an outlier if its density is relatively **much lower** than that of its neighbors

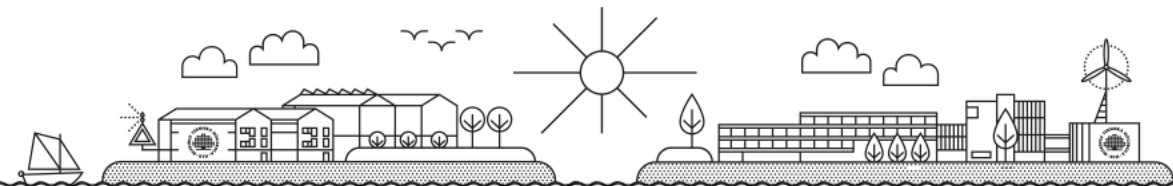


DISTANCE-BASED OUTLIER DETECTION (1)

- For each object o , examine the number of other objects in the r -neighborhood of o , where $r > 0$ is a user-specified **distance threshold**
- An object o is an outlier if most ($0 < \pi \leq 1$ as a **fraction threshold**) of the objects in D are far away from o , i.e., not in the r -neighborhood of o

- An object o is a $DB(r, \pi)$ -outlier if
$$\frac{|\{o' | dist(o, o') \leq r\}|}{|D|} \leq \pi$$

- Equivalently, one can check the distance between o and its k -th nearest neighbor o_k , where $k = \lceil \pi |D| \rceil$. o is an outlier if $dist(o, o_k) > r$



DISTANCE-BASED OUTLIER DETECTION (2)

○ Efficient computation: Nested loop algorithm

Algorithm: Distance-based outlier detection.

Input:

- a set of objects $D = \{o_1, \dots, o_n\}$, threshold r ($r > 0$) and π ($0 < \pi \leq 1$);

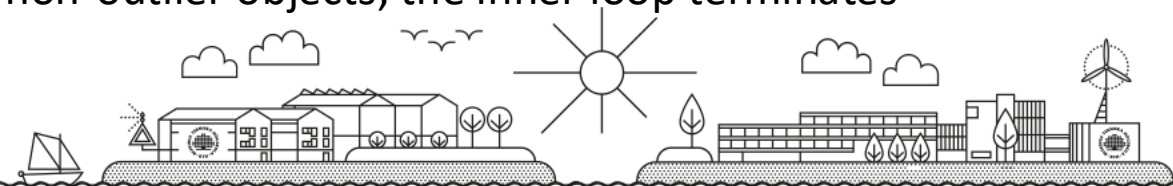
Output: $DB(r, \pi)$ outliers in D .

Method:

```
for i = 1 to n do
  count ← 0
  for j = 1 to n do
    if i ≠ j and dist(oi, oj) ≤ r then
      count ← count + 1
      if count ≥ π · n then
        exit {oi cannot be a DB(r, π) outlier}
      endif
    endif
  endfor
  print oi {oi is a DB(r, π) }
endfor;
```

$$\frac{|\{o' | dist(o, o') \leq r\}|}{|D|} \leq \pi$$

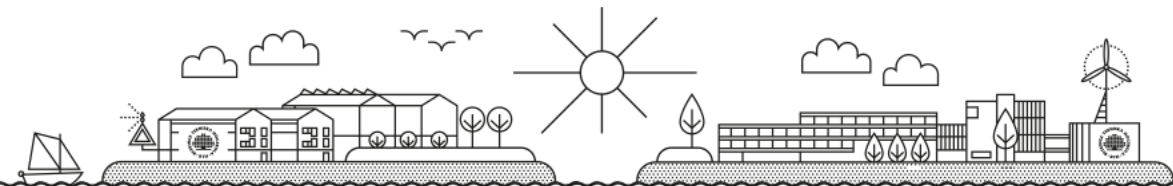
Efficiency: CPU time is not $O(n^2)$ but linear to the data set size since for most non-outlier objects, the inner loop terminates early



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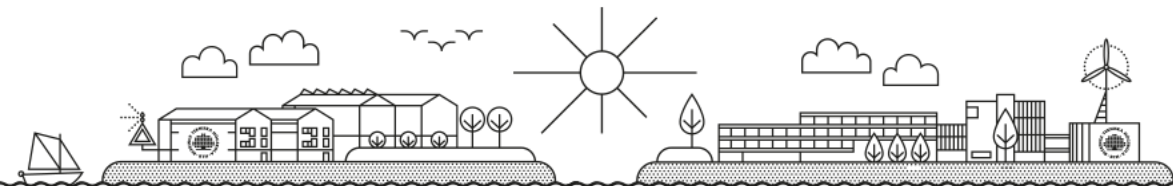
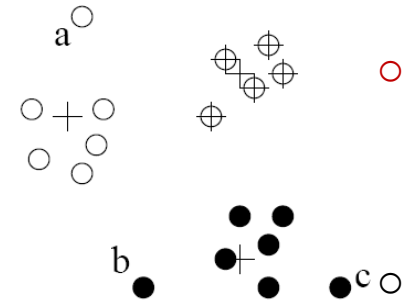
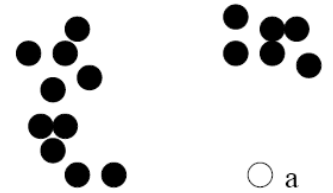


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CLUSTERING-BASED OUTLIER DETECTION (1)

- An object is an outlier if **(1)** it does **not belong** to any cluster, **(2)** there is a **large distance** between the object and its closest cluster, or **(3)** it belongs to a **small or sparse** cluster
- **Case 1:** Not belong to any cluster
 - Identify animals not part of a flock: Using a density-based clustering method such as DBSCAN
- **Case 2:** Far from its closest cluster
 - Using k-means, partition data points into k clusters
 - For each object o, assign an outlier score based on its distance from its closest center
 - If $\text{dist}(o, C_o) / \text{avg_dist}(C_o)$ is large, likely an outlier
- E.g., Intrusion detection: Consider the similarity between data points and the clusters in a training data set
 - Use a training set to find patterns of “**normal**” data, e.g., Daily TCP connections, frequent itemsets in each segment, and cluster similar connections into groups
 - Compare new data points with the clusters mined—Outliers are possible attacks



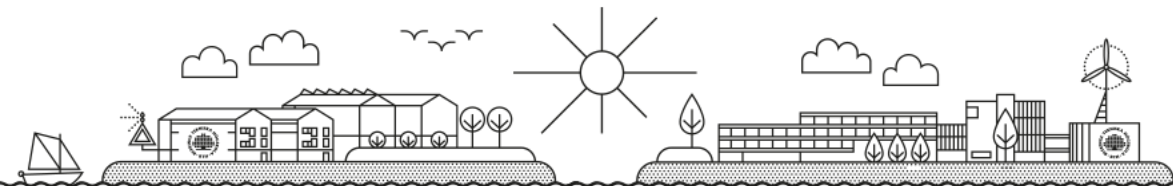
CLUSTERING-BASED METHOD: STRENGTH AND WEAKNESS

○ Strength

- Detect outliers without requiring any labeled data
- Work for many types of data
- Clusters can be regarded as summaries of the data
- Once the cluster are obtained, need only compare any object against the clusters to determine whether it is an outlier (fast)

○ Weakness

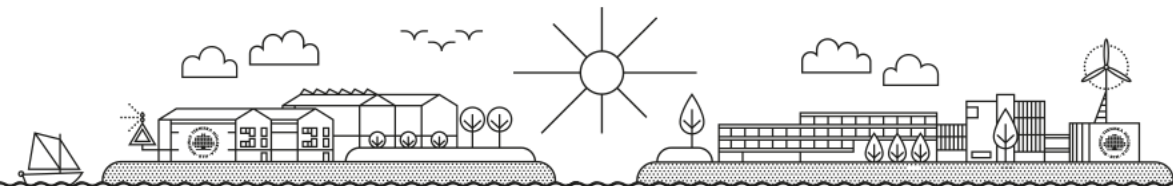
- Effectiveness depends highly on the clustering method used—they may not be optimized for outlier detection
- High computational cost: Need to first find clusters



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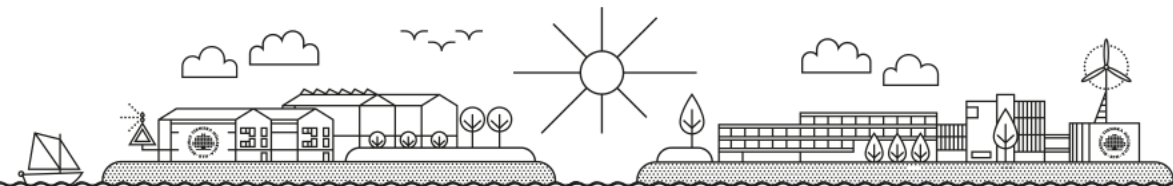
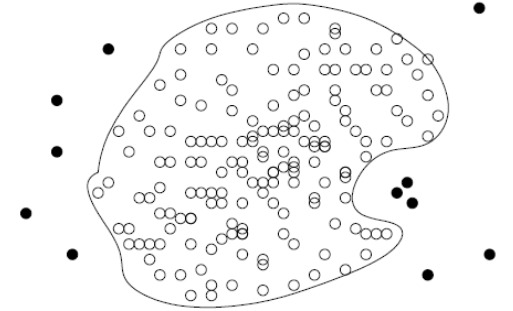


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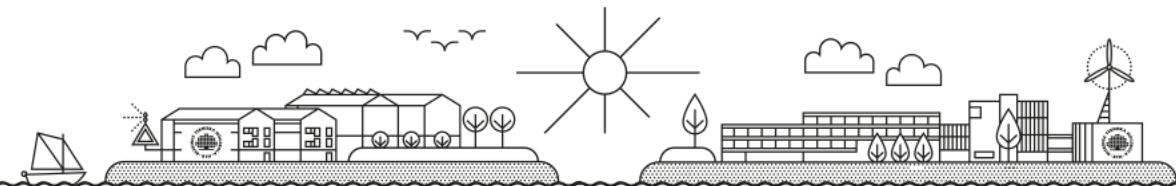
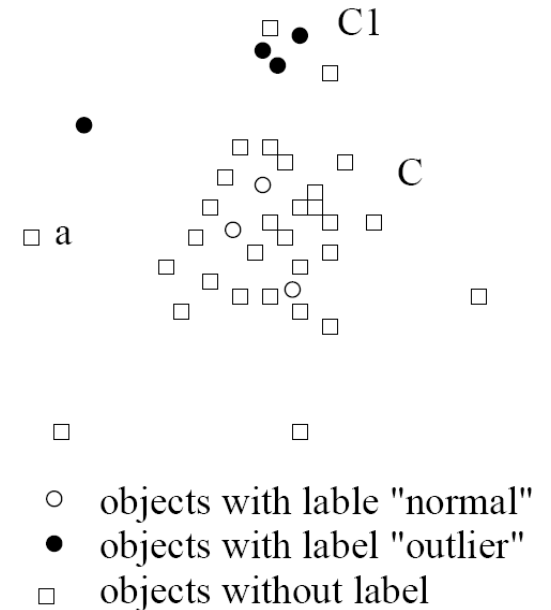
CLASSIFICATION-BASED METHOD (1): ONE-CLASS MODEL

- **Idea:** Train a classification model that can distinguish “normal” data from outliers
- **A brute-force approach:** Consider a training set that contains samples labeled as “normal” and others labeled as “outliers”
 - But, the training set is typically heavily biased: # of “normal” samples likely far exceeds number of outlier samples
 - Cannot detect unseen anomaly
- **One-class model:** A classifier is built to describe only the normal class.
 - Learn the decision boundary of the normal class using classification methods such as SVM
 - Any samples that do not belong to the normal class (not within the decision boundary) are declared as outliers
 - **Advantage:** can detect new outliers that may not appear close to any outlier objects in the training set
 - Extension: Normal objects may belong to multiple classes



CLASSIFICATION-BASED METHOD (2): SEMI-SUPERVISED LEARNING

- **Semi-supervised learning:** Combining classification-based and clustering-based methods
- **Method**
 1. Using a clustering-based approach, find a **large cluster**, C , and a **small cluster**, C_1
 2. Since **some objects in C** carry the label “normal”, treat **all** objects in C as normal
 3. Use the one-class model of this cluster to identify normal objects in outlier detection
 4. Since **some objects in C_1** carry the label “outlier”, declare **all** objects in C_1 as outliers
 5. Any object that does not fall into the model for C (such as a) is considered an outlier as well
- Comments on classification-based outlier detection methods
 - **Strength:** Outlier detection is fast
 - **Bottleneck:** Quality heavily depends on the availability and quality of the training set, but often difficult to obtain representative and high-quality training data



REFERENCES



- [1]. Han, Jiawei, Jian Pei, and Hanghang Tong. *Data mining: concepts and techniques*. Morgan kaufmann, 2022.

